A Hardy space approximation supporting zero-free half-planes for the ζ-function

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Abstract

An equivalent version of the Báez-Duarte criterion [2] for the Riemann Hypothesis (RH) by Bagchi [5] states that the RH holds true if and only if the function $E(s) = 1/s$ belongs to the closed linear span of $G_k(s) = (k^{-s} - k^{-1})\zeta(s)/s$, $k \geq 2$ in the Hardy space $H^2(\mathbb{C}_{1/2})$, where $\mathbb{C}_\alpha$ denotes the half-plane $\mathrm{Re}(s) > \alpha$. We first show that if E belongs to the closure of $\mathrm{span}(G_k)_{k\geq 2}$ in $H^2(\mathbb{C}_\alpha)$ for $\alpha > 1/2$, then ζ is zero-free in $\mathbb{C}_\alpha$. We then use this as the basis for a numerical analysis of the sequence

$$s_n = \left\| \sum_{k=2}^{n} \mu(k) G_k - E \right\|_\alpha^2,$$

for $1/2 \leq \alpha \leq 1$, where $\|.\|_\alpha$ is the norm in $H^2(\mathbb{C}_\alpha)$ and μ the Möbius function.

Keywords: Riemann hypothesis, zero-free half-planes, Báez-Duarte criterion, Hardy spaces, numerical approximation.

1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of the Riemann zeta function lie on the critical line $\mathrm{Re}(s) = \frac{1}{2}$. Its implications for the distribution of prime numbers make it one of the central open problems in mathematics.

A functional-analytic approach to RH was initiated by Nyman and Beurling, who showed that RH is equivalent to the density of a specific family of functions in $L^2(0,1)$ [6, 11]. Báez-Duarte later refined this by replacing the original uncountable family with a countable sequence, simplifying the problem while preserving its essential structure [2]. Balazard and Saias extended

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this approach to a weighted ℓ^2-sequence space by posing the question of whether the orthogonal projection coefficients $c_{n,k}$ of $\mathbf{1}$ onto $\operatorname{span}\{r_k : 2 \le k \le n\}$ satisfy

$$\lim_{n\to\infty} c_{n,k} = -\frac{\mu(k)}{k}, \quad \forall\, k \ge 2,$$

where μ is the Möbius function and $r_k(j) = j \bmod k$ denotes the remainder upon division of j by k. This condition was shown by Weingartner in [13] to be equivalent to the RH. This suggests that the choice of coefficients $\mu(k)/k$ (or $\mu(k)$ in our situation below after normalizing) is in a strict sense the best possible one.

A reformulation within the Hardy space $H^2(\mathbb{C}_{1/2})$ was established by Bagchi [5], where $\mathbb{C}_\alpha = \{s \in \mathbb{C} : \operatorname{Re}(s) > \alpha\}$ for $\alpha \in \mathbb{R}$: RH holds if and only if $E(s) = 1/s$ belongs to the closed linear span of

$$G_k(s) = \left(k^{-s} - k^{-1}\right)\frac{\zeta(s)}{s}, \quad k \ge 1,$$

in $H^2(\mathbb{C}_{1/2})$. Building on these ideas, Noor introduced an equivalent reformulation in the Hardy–Hilbert space $H^2(\mathbb{D})$ of the unit disk [10]: RH holds if and only if the constant function $\mathbf{1}$ belongs to the closed linear span of

$$h_k(z) = \frac{1}{1-z}\log\left(\frac{1+z+\cdots+z^{k-1}}{k}\right), \quad k \ge 2.$$

In the same work, Noor established that the series $\sum_{k=2}^{\infty}\frac{\mu(k)}{k}(I-S)h_k$ converges to $1-z$ in $H^2(\mathbb{D})$, where S is the shift operator. For an overview of the various equivalent reformulations of the Nyman-Beurling and Báez-Duarte approaches that appear in the literature, see [7, p. 5-6].

The main idea of this paper is to generalize this result in an attempt to approximate RH using the reformulation of Bagchi in $H^2(\mathbb{C}_{1/2})$. To achieve this, we employ the Hilbert space of Dirichlet series $\mathcal{H}^2$, which presents a natural connection with the Hardy space of the disk. Within this framework, we construct a bounded operator $\mathcal{M}^\tau$ that establishes a connection between $\mathcal{H}^2$ and the Hardy spaces $H^2(\mathbb{C}_\alpha)$, leading to our main result (Theorem 3.1): the series $\sum_{k=2}^{\infty}\mu(k)G_k$ converges to E in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$. Furthermore, Proposition 3.1 establishes that for any $\alpha \ge 1/2$,

$$E \in \overline{\operatorname{span}}\{G_k : k \ge 2\} \text{ in } H^2(\mathbb{C}_\alpha) \implies \zeta(s) \ne 0, \quad \forall\, s \in \mathbb{C}_\alpha.$$

An important question that arises is whether the convergence of the series extends to $1/2 \le \alpha \le 1$. Báez-Duarte [3] showed that the L^2-analog of the sequence s_n diverges in $L^2(0,1)$, which implies the divergence of s_n itself in $H^2(\mathbb{C}_{1/2})$. However, this does not discount the possibility of some subsequence of s_n converging to 0. In fact, the numerical evidence we present below strongly suggests that such subsequences exist. Studying the convergence in the range $1/2 < \alpha \le 1$ is therefore essential, as it could provide information about zero-free regions of ζ within the critical strip, a problem that remains open and is widely considered to be as difficult as the RH itself.

The numerical analysis conducted in this work investigates the behavior of the partial sums

$$s_n = \left\|\sum_{k=2}^{n}\mu(k)G_k - E\right\|^2_{H^2(\mathbb{C}_\alpha)}$$

for $0.5 \leq \alpha \leq 1.0$. For α close to 1, the sequence s_n exhibits a clear decreasing trend, suggesting convergence in this region. For α close to 0.5, the sequence shows persistent oscillations and significant deviations, indicating divergence consistent with the theoretical result of Báez-Duarte. Section 2 introduces the necessary function spaces and key results. Section 3 develops the main theoretical framework. Section 4 presents the numerical experiments and their implications.

2 Preliminaries

We collect the function spaces and key results needed throughout the paper. The three Hilbert spaces $H^2(\mathbb{D})$, $\mathcal{H}^2$, and $H^2(\mathbb{C}_{1/2})$ are introduced here as the analytic framework within which the completeness conditions equivalent to the Riemann Hypothesis are formulated and studied.

2.1 Hardy–Hilbert spaces and Dirichlet series

Let $H^2(\mathbb{D})$ be the Hardy–Hilbert space of the unit disk, defined as

$$H^2(\mathbb{D}) = \left\{ f(z) = \sum_{n=0}^{\infty} a_n z^n : \|f\|_{H^2}^2 = \sum_{n=0}^{\infty} |a_n|^2 < \infty \right\}.$$

A recent reformulation of the RH, introduced in [10], is stated as follows. For each $k \geq 2$, define

$$h_k(z) = \frac{1}{1-z} \log\left(\frac{1 + z + \cdots + z^{k-1}}{k} \right).$$

Theorem 2.1 (Theorem 6, [10]). *The Riemann Hypothesis holds if and only if the constant function* $\mathbf{1}$ *belongs to the closed linear span of* $\{h_k : k \geq 2\}$ *in* $H^2(\mathbb{D})$.

Let $S = M_z$ denote the shift operator on $H^2(\mathbb{D})$, corresponding to multiplication by z. An additional result from [10] plays a central role in this paper.

Lemma 2.1 (Lemma 11, [10]). *The series* $\sum_{k=2}^{\infty} \frac{\mu(k)}{k}(I - S)h_k$ *converges to* $1 - z$ *in* $H^2(\mathbb{D})$, *where* μ *is the Möbius function.*

Let $\mathcal{H}^2$ denote the Hilbert space of Dirichlet series, defined as

$$\mathcal{H}^2 = \left\{ f(s) = \sum_{n=1}^{\infty} a_n n^{-s} : \|f\|_{\mathcal{H}^2}^2 = \sum_{n=1}^{\infty} |a_n|^2 < \infty \right\}.$$

Setting $H_0^2(\mathbb{D}) = H^2(\mathbb{D}) \ominus \mathbb{C}$, there exists an isometric isomorphism

$$\psi : H_0^2(\mathbb{D}) \longrightarrow \mathcal{H}^2, \qquad z^k \longmapsto k^{-s}.$$

2.2 The Hardy space of $\mathbb{C}_{1/2}$

For $\alpha \in \mathbb{R}$, let $\mathbb{C}_\alpha = \{s = \sigma + it : \sigma > \alpha\}$ and let $H^2(\mathbb{C}_\alpha)$ denote the Hardy space of analytic functions F on $\mathbb{C}_\alpha$ satisfying

$$\|F\|^2_{H^2(\mathbb{C}_\alpha)} := \sup_{x>\alpha} \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(x+it)|^2 \, dt < \infty.$$

Any such F admits a non-tangential boundary value F^* almost everywhere on $\{\mathrm{Re}(s) = \alpha\}$, with

$$\|F\|^2_{H^2(\mathbb{C}_\alpha)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F^*(\alpha+it)|^2 \, dt.$$

Moreover, $H^2(\mathbb{C}_\alpha)$ is a Hilbert space with reproducing kernel

$$k_w(s) = \frac{1}{s + \overline{w} - 2\alpha}, \quad w \in \mathbb{C}_\alpha;$$

in particular, convergence in norm implies pointwise convergence in $\mathbb{C}_\alpha$.

Define the functions G_k and E in $H^2(\mathbb{C}_{1/2})$, for $k \geq 2$, as

$$G_k(s) = (k^{-s} - k^{-1}) \frac{\zeta(s)}{s} \quad \text{and} \quad E(s) = \frac{1}{s}.$$

The Riemann Hypothesis reformulation for $H^2(\mathbb{C}_{1/2})$ can be stated as follows.

Theorem 2.2 (Theorem 2, [5]). *RH is true if and only if E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_{1/2})$.*

3 An $H^2(\mathbb{C}_\alpha)$ approximation approach

The main goal of this section is to show that E belongs to the closed linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_\alpha)$, for every $\alpha > 1$. For $\tau > 1/2$ and $k \geq 2$, define

$$g_{k,\tau}(z) = k^{\tau-1} \sum_{n=1}^{\infty} \frac{z^n}{n^\tau} - \sum_{n=1}^{\infty} \frac{z^{nk}}{n^\tau}.$$

It is clear that $g_{k,\tau} \in H^2_0$. Furthermore, we observe that

$$\begin{aligned}
(\psi g_{k,\tau})(s) &= k^{\tau-1} \sum_{n=1}^{\infty} \frac{n^{-s}}{n^\tau} - k^{-s} \sum_{n=1}^{\infty} \frac{n^{-s}}{n^\tau} \\
&= (k^{\tau-1} - k^{-s})\zeta(s+\tau) \\
&= k^{\tau-1} f_{k,\tau} \quad \text{for } \mathrm{Re}(s) > 1/2,
\end{aligned}$$

where $f_{k,\tau}(s) = (1 - k^{1-s-\tau})\zeta(s+\tau)$.

3.1 A bounded operator from $\mathcal{H}^2$ to $H^2(\mathbb{C}_\alpha)$

The operator $\mathcal{M}_{1/s} : \mathcal{H}^2 \to H^2(\mathbb{C}_{1/2})$, defined by $\mathcal{M}_{1/s} : f(s) \mapsto f(s)/s$, is bounded according to [12]. For $\tau \geq 1/2$, we extend this construction by defining $\mathcal{M}^\tau : \mathcal{H}^2 \to H^2(\mathbb{C}_{1/2+\tau})$ via $\mathcal{M}^\tau : f \longmapsto f(s-\tau)/s$. To see that $\mathcal{M}^\tau$ is well defined and bounded, note that for $\sigma > \tau + 1/2$,

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{|f(\sigma-\tau+it)|^2}{|\sigma+it|^2}\,dt \leq \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{|f(\sigma-\tau+it)|^2}{|\sigma-\tau+it|^2}\,dt = \|\mathcal{M}_{1/s}f\|^2_{H^2(\mathbb{C}_{1/2})},$$

and taking the supremum over $\sigma > \tau + 1/2$ yields

$$\|\mathcal{M}^\tau f\|_{H^2(\mathbb{C}_{1/2+\tau})} \leq \|\mathcal{M}_{1/s}\|\,\|f\|_{\mathcal{H}^2}, \qquad \forall f \in \mathcal{H}^2.$$

The relevance of $\mathcal{M}^\tau$ to our approach lies in how it acts on the functions 1 and $f_{k,\tau}$. A direct computation gives

$$(\mathcal{M}^\tau 1)(s) = \frac{1}{s} = E(s),$$

and

$$(\mathcal{M}^\tau f_{k,\tau})(s) = \frac{f_{k,\tau}(s-\tau)}{s} = (1-k^{1-s})\frac{\zeta(s)}{s} = -kG_k(s).$$

3.2 Zero free region of ζ in $\mathbb{C}_\alpha$

We now establish one of the main results of the paper. The key step is the following generalization of Lemma 2.1 to the family $\{g_{k,\tau}\}$.

Lemma 3.1. *The series $\sum_{k=2}^{\infty} \frac{\mu(k)}{k^\tau} g_{k,\tau}$ converges to $-z$ in H_0^2, for each $\tau > 1/2$.*

Proof. Note that

$$\begin{aligned}\sum_{k=2}^{n} \frac{\mu(k)}{k^\tau} g_{k,\tau}(z) &= \sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \left\{ k^{\tau-1}\sum_{j=1}^{\infty}\frac{z^j}{j^\tau} - \sum_{j=1}^{\infty}\frac{z^{jk}}{j^\tau} \right\} \\ &= \sum_{k=1}^{n} \frac{\mu(k)}{k} \sum_{j=1}^{\infty}\frac{z^j}{j^\tau} - \sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty}\frac{z^{jk}}{j^\tau}.\end{aligned}$$

Since $\sum_{k=1}^{\infty} \frac{\mu(k)}{k} = 0$, the first term vanishes in the limit:

$$\left\| \sum_{k=1}^{n} \frac{\mu(k)}{k} \sum_{j=1}^{\infty}\frac{z^j}{j^\tau} \right\|_{H_0^2} = \left| \sum_{k=1}^{n} \frac{\mu(k)}{k} \right| \left\| \sum_{j=1}^{\infty}\frac{z^j}{j^\tau} \right\|_{H_0^2} \longrightarrow 0.$$

It therefore suffices to show that

$$\left\| -\sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty}\frac{z^{jk}}{j^\tau} + z \right\|_{H_0^2} \longrightarrow 0. \tag{3.1}$$

A rearrangement of the double sum gives

$$\begin{aligned}
\sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty} \frac{z^{jk}}{j^\tau} &= \sum_{j=1}^{n} \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j,\\ 1\le d\le n}} \mu(d) \right) + \sum_{j=n+1}^{\infty} \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j,\\ 1\le d\le n}} \mu(d) \right) \\
&= \sum_{j=1}^{n} \frac{z^j}{j^\tau} \left[\frac{1}{j} \right] + \sum_{j=n+1}^{\infty} \frac{z^j}{j^\tau} \left(\sum_{\substack{d|j,\\ 1\le d\le n}} \mu(d) \right), \qquad (3.2)
\end{aligned}$$

where the last equality uses the identity $\sum_{d|j} \mu(d) = \left[\frac{1}{j}\right]$. Returning to (3.2),

$$\sum_{k=1}^{n} \frac{\mu(k)}{k^\tau} \sum_{j=1}^{\infty} \frac{z^{jk}}{j^\tau} = z + \phi_n(z),$$

where

$$\phi_n(z) = \sum_{j=n+1}^{\infty} \left(\frac{1}{j^\tau} \sum_{\substack{d|j,\\ 1\le d\le n}} \mu(d) \right) z^j.$$

It remains to show that $\|\phi_n\|_{H_0^2} \to 0$ as $n \to \infty$. Letting $\sigma(n)$ denote the number of divisors of n, we estimate

$$\|\phi_n\|_{H_0^2}^2 = \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau}} \left| \sum_{\substack{d|j,\\ 1\le d\le n}} \mu(d) \right|^2 \le \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau}} \left(\sum_{d|j} 1 \right)^2 = \sum_{j=n+1}^{\infty} \frac{\sigma(j)^2}{j^{2\tau}}.$$

Since $\sigma(n) = o(n^\epsilon)$ for every $\epsilon > 0$, choosing $\epsilon > 0$ such that $\tau > 1/2 + \epsilon$ yields

$$\|\phi_n\|_{H_0^2}^2 \lesssim \sum_{j=n+1}^{\infty} \frac{j^{2\epsilon}}{j^{2\tau}} = \sum_{j=n+1}^{\infty} \frac{1}{j^{2\tau-2\epsilon}} \longrightarrow 0,$$

since $2\tau - 2\epsilon > 1$. □

Theorem 3.1. *The series* $\sum_{k=2}^{\infty} \mu(k) G_k$ *converges to* E *in* $H^2(\mathbb{C}_{1/2+\tau})$, *for every* $\tau > 1/2$.

Proof. By Lemma 3.1 and the isomorphism ψ,

$$\sum_{k=2}^{n} \frac{\mu(k)}{k} f_{k,\tau} \longrightarrow -1 \quad \text{in } \mathcal{H}^2.$$

Applying $\mathcal{M}^\tau$ and using $(\mathcal{M}^\tau f_{k,\tau})(s) = -kG_k(s)$, we conclude

$$\sum_{k=2}^{n} \mu(k) G_k \longrightarrow E \quad \text{in } H^2(\mathbb{C}_{1/2+\tau}). \qquad (3.3)$$

□

Corollary 3.1. *E belongs to the closure linear span of $\{G_k : k \geq 2\}$ in $H^2(\mathbb{C}_\alpha)$, for every $\alpha > 1$.*

As another consequence, Theorem 3.1 yields a new proof of the non-vanishing of ζ in the half-plane $\mathbb{C}_1$.

Proposition 3.1. *For any $\alpha \geq 1/2$,*

$$E \in \overline{\operatorname{span}}\{G_k : k \geq 2\} \text{ in } H^2(\mathbb{C}_\alpha) \implies \zeta(s) \neq 0, \quad \forall s \in \mathbb{C}_\alpha.$$

Proof. Let ρ be a zero of ζ in $\mathbb{C}_\alpha$. Then $G_k(\rho) = 0$ for every $k \geq 2$. Since $H^2(\mathbb{C}_\alpha)$ is a reproducing kernel Hilbert space, convergence in norm implies pointwise convergence, so $E(\rho) = 0$, a contradiction. □

4 Numerical evidence

Theorem 3.1 establishes that $\sum \mu(k)G_k$ converges to E in $H^2(\mathbb{C}_\alpha)$ for every $\alpha > 1$, while Proposition 3.1 shows that the membership of E in the closed linear span of $\{G_k : k \geq 2\}$ forces ζ to be zero-free on $\mathbb{C}_\alpha$. The convergence is settled for $\alpha > 1$; the range $1/2 \leq \alpha \leq 1$ (precisely the portion of the critical strip where the behaviour of ζ is least understood) lies beyond the reach of our analytical methods. The purpose of this section is to probe this range numerically.

We study the partial sums $S_n = \sum_{k=2}^{n} \mu(k)G_k \in H^2(\mathbb{C}_\alpha)$, and monitor the approximation error $s_n = \|S_n - E\|^2_{H^2(\mathbb{C}_\alpha)}$. By Proposition 3.1, numerical evidence that $s_n \to 0$ for a given α is evidence in favour of the non-vanishing of ζ on $\mathbb{C}_\alpha$. At the opposite extreme, the $L^2(0,1)$-divergence established by Báez-Duarte [3] propagates to s_n at $\alpha = 1/2$, so that genuine convergence cannot be expected on the critical line itself. Our experiments are accordingly designed to locate, as a function of α, the transition between these two regimes.

For a fixed α, each error s_n is obtained by numerically evaluating the boundary integral that defines the $H^2(\mathbb{C}_\alpha)$ norm of $S_n - E$. This integral was computed by adaptive Gauss–Kronrod quadrature (the QUADPACK routine exposed through `scipy.integrate.quad`) in IEEE double precision, with the Riemann zeta function obtained from its analytic continuation as implemented in `scipy.special.zeta`. Starting from S_2, the partial sum was extended one term at a time and the integral recomputed after each increment, the iteration halting as soon as $s_n < \varepsilon$ or the prescribed time budget was exhausted.

All computations were performed in Python 3.13 on a machine with a 5.2 GHz Intel Core i9-12900K processor, 128 GB of DDR4 RAM, running Ubuntu 22.04 LTS.

4.1 Experiments with tolerance and time

Our first experiment quantifies how the rate of convergence of (S_n) degrades as α approaches the critical line. We discretised the interval $[0.5, 1.0]$ into 100 equally spaced points,

$$\alpha_i = 0.5 + 0.005\,(i - 1), \qquad i = 1, \ldots, 100,$$

and, for three tolerance levels $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, recorded for each α_i the smallest number of terms n for which $s_n < \varepsilon$, together with the time required to reach it. Here ε fixes a prescribed accuracy of approximation, while n serves as an intrinsic, machine-independent measure of how rapidly S_n approaches E: the larger the n needed to meet a fixed ε, the slower the convergence. To keep the computation feasible, each run was capped at a time budget of 3600 seconds; when this limit was reached, the run was terminated and the largest n attained was recorded. The results are reported in Figure 1 and Table 1.

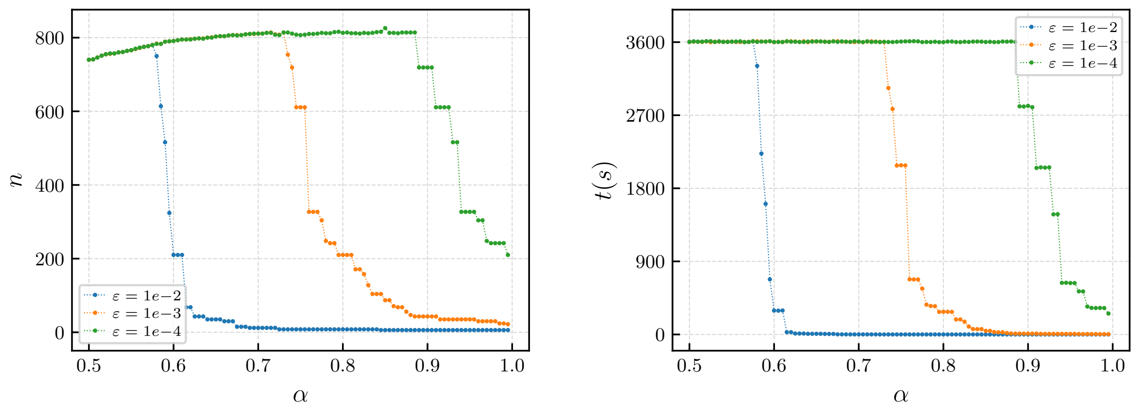

Figure 1: Tolerance test

α	$\varepsilon = 1\text{e}{-2}$		$\varepsilon = 1\text{e}{-3}$		$\varepsilon = 1\text{e}{-4}$	
	n	Time (s)	n	Time (s)	n	Time (s)
0.50	740	3.600e+3	740	3.600e+3	740	3.600e+3
0.55	766	3.600e+3	766	3.600e+3	766	3.600e+3
0.60	210	2.941e+2	791	3.600e+3	791	3.600e+3
0.65	35	8.237e+0	803	3.600e+3	803	3.600e+3
0.70	12	8.330e−1	811	3.600e+3	811	3.600e+3
0.75	8	3.520e−1	611	2.084e+3	807	3.600e+3
0.80	8	3.500e−1	210	2.792e+2	813	3.600e+3
0.85	6	1.760e−1	87	4.466e+1	826	3.600e+3
0.90	6	1.760e−1	43	1.057e+1	719	2.815e+3
0.95	6	1.760e−1	35	6.988e+0	327	6.333e+2

Table 1: Corresponding to Figure 1.

A sharp, tolerance-dependent transition is evident. The coarse target $\varepsilon = 10^{-2}$ is met for every $\alpha \geq 0.60$, requiring only six to twelve terms once $\alpha \geq 0.70$. The intermediate target $\varepsilon = 10^{-3}$ is met for $\alpha \geq 0.75$, and the fine target $\varepsilon = 10^{-4}$ only for $\alpha \geq 0.90$; thus the threshold in α below which the time budget is exhausted rises steadily as the demanded accuracy increases. Below this transition the cost escalates abruptly, and for $\alpha \leq 0.55$ none of the three tolerances is reached within the allotted 3600 seconds.

Two features of this transition deserve emphasis. First, Theorem 3.1 guarantees convergence of S_n to E only for $\alpha > 1$, yet the experiments drive the error below 10^{-3} for α as small as ≈ 0.75. The analytic threshold $\alpha > 1$ is therefore not sharp, and the data support the expectation that convergence persists well into the critical strip. Second, the failure to attain a prescribed tolerance at small α must be read with care: it records a convergence rate that deteriorates rapidly as $\alpha \to 1/2$, and is consistent with divergence. Disentangling these two readings near the endpoints is the subject of the next subsection.

4.2 Case studies at the endpoints $\alpha \approx 1/2$ and $\alpha \approx 1$

The transition identified in Subsection 4.1 separates two qualitatively distinct regimes. To examine each in isolation, we refine the analysis near the two endpoints of the interval. For $\alpha_{\min} \in \{0.5, 0.9\}$ we evaluate s_n on the grids

$$\alpha_i = \alpha_{\min} + 0.01\, i, \qquad i = 0, 1, \ldots, 9,$$

at the sample points $n = 1000\, j$, $j = 1, \ldots, 100$, so that the behaviour of s_n is tracked up to $n = 10^5$. This larger range of n lets the asymptotic trend of each sequence emerge, beyond the onset captured by the tolerance experiment of the previous subsection.

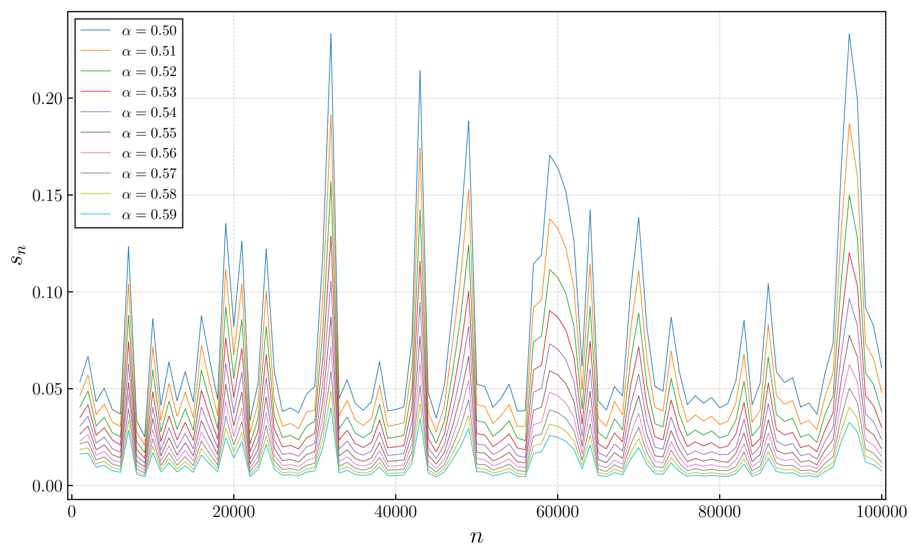

Figure 2: $0.50 \leq \alpha \leq 0.59$

The divergent regime $0.50 \leq \alpha \leq 0.59$**.** Figure 2 and Table 2 show that, across this entire interval, s_n exhibits prominent oscillations with no stabilising trend. Not only does the sequence fail to decay, but even its local minima remain bounded away from zero: at $\alpha = 0.50$ they hover near 4×10^{-2} while the peaks reach order 10^{-1}, recurring without attenuation up to $n = 10^5$. This is precisely the behaviour anticipated by theory. Báez-Duarte [3] proved that the $L^2(0,1)$ analogue of s_n diverges, and this divergence transfers to $H^2(\mathbb{C}_{1/2})$, so that $s_n \not\to 0$ at $\alpha = 1/2$. The numerics show that this obstruction is robust throughout the lower edge of the strip, and not confined to the critical line itself.

n	s_n									
	$\alpha = 0.50$	$\alpha = 0.51$	$\alpha = 0.52$	$\alpha = 0.53$	$\alpha = 0.54$	$\alpha = 0.55$	$\alpha = 0.56$	$\alpha = 0.57$	$\alpha = 0.58$	$\alpha = 0.59$
1000	5.358e−2	4.651e−2	4.039e−2	3.510e−2	3.050e−2	2.655e−2	2.313e−2	2.155e−2	1.870e−2	1.625e−2
10000	8.618e−2	7.170e−2	5.968e−2	4.836e−2	4.037e−2	3.461e−2	2.891e−2	2.417e−2	2.022e−2	1.692e−2
20000	8.215e−2	6.742e−2	5.537e−2	4.549e−2	3.739e−2	3.077e−2	2.534e−2	2.089e−2	1.723e−2	1.421e−2
30000	5.103e−2	3.865e−2	3.322e−2	2.682e−2	2.169e−2	1.753e−2	1.419e−2	1.149e−2	9.308e−3	7.468e−3
40000	3.926e−2	3.131e−2	2.498e−2	1.994e−2	1.592e−2	1.272e−2	1.016e−2	8.060e−3	6.438e−3	5.149e−3
50000	5.233e−2	4.180e−2	3.343e−2	2.674e−2	2.143e−2	1.716e−2	1.376e−2	1.103e−2	8.851e−3	7.112e−3
60000	1.642e−1	1.328e−1	1.075e−1	8.699e−2	7.050e−2	5.715e−2	4.633e−2	3.762e−2	3.057e−2	2.486e−2
70000	1.384e−1	1.110e−1	8.912e−2	7.162e−2	5.758e−2	4.633e−2	3.731e−2	3.009e−2	2.428e−2	1.960e−2
80000	4.016e−2	3.152e−2	2.461e−2	1.933e−2	1.598e−2	1.253e−2	9.831e−3	7.715e−3	6.061e−3	4.762e−3
90000	4.038e−2	3.171e−2	2.490e−2	1.955e−2	1.537e−2	1.208e−2	9.757e−3	7.671e−3	6.034e−3	4.750e−3
100000	6.073e−2	4.805e−2	3.805e−2	2.996e−2	2.377e−2	1.887e−2	1.508e−2	1.191e−2	9.475e−3	7.543e−3

Table 2: Corresponding to Figure 2.

The convergent regime $0.90 \le \alpha \le 0.99$**.** Figure 3 and Table 3 reveal a markedly different picture. The sequence s_n now displays a clear overall decay, with oscillatory peaks whose amplitude diminishes steadily as n grows; although the decrease is not strictly monotone, the envelope of the sequence contracts consistently. For $\alpha = 0.99$, the error falls from 1.52×10^{-4} at $n = 10^3$ to 3.13×10^{-6} at $n = 10^5$, a reduction of nearly two orders of magnitude, and the same qualitative behaviour holds uniformly across the interval. These observations strongly suggest that $S_n \to E$ in $H^2(\mathbb{C}_\alpha)$ throughout $0.90 \le \alpha \le 0.99$.

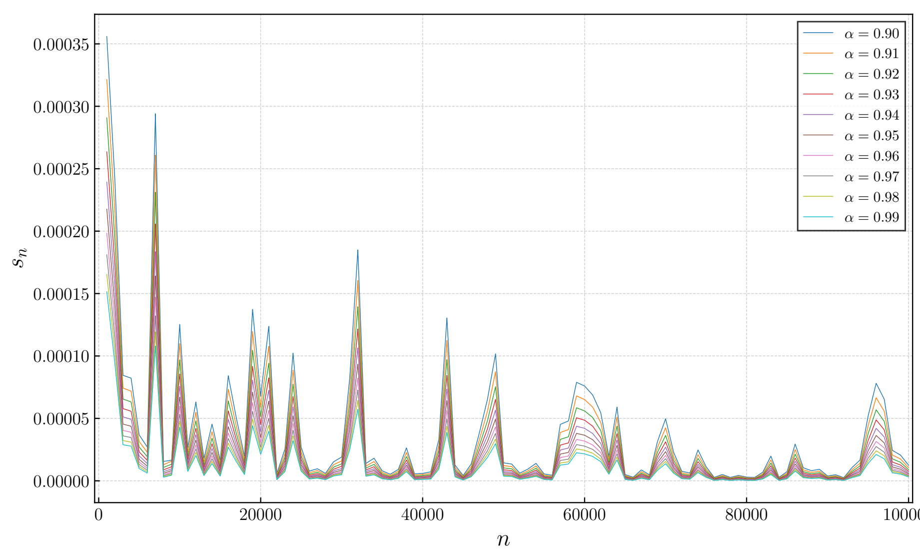

Figure 3: $0.90 \le \alpha \le 0.99$

We stress the conditional nature of this reading. By Proposition 3.1, genuine convergence $S_n \to E$ for such α would place E in the closed span of $\{G_k : k \ge 2\}$ and hence furnish a zero-free half-plane $\mathbb{C}_\alpha$ deep within the critical strip, a conclusion lying far beyond all currently known zero-free regions and widely regarded as being as difficult as the Riemann Hypothesis itself. The evidence presented here is therefore numerical and suggestive, not a proof.

n	s_n									
	$\alpha = 0.90$	$\alpha = 0.91$	$\alpha = 0.92$	$\alpha = 0.93$	$\alpha = 0.94$	$\alpha = 0.95$	$\alpha = 0.96$	$\alpha = 0.97$	$\alpha = 0.98$	$\alpha = 0.99$
1000	3.558e−4	3.215e−4	2.909e−4	2.636e−4	2.392e−4	2.177e−4	1.983e−4	1.812e−4	1.654e−4	1.516e−4
10000	1.253e−4	1.100e−4	9.691e−5	8.562e−5	7.583e−5	6.707e−5	5.974e−5	5.335e−5	4.775e−5	4.297e−5
20000	6.772e−5	5.882e−5	5.149e−5	4.498e−5	3.939e−5	3.463e−5	3.051e−5	2.699e−5	2.395e−5	2.133e−5
30000	1.905e−5	1.594e−5	1.360e−5	1.164e−5	9.845e−6	8.498e−6	7.364e−6	6.408e−6	5.598e−6	4.909e−6
40000	5.966e−6	5.138e−6	4.214e−6	3.460e−6	2.849e−6	2.350e−6	1.945e−6	1.616e−6	1.347e−6	1.127e−6
50000	1.437e−5	1.215e−5	1.032e−5	8.802e−6	7.528e−6	6.473e−6	5.585e−6	4.839e−6	4.212e−6	3.683e−6
60000	7.597e−5	6.516e−5	5.604e−5	4.854e−5	4.202e−5	3.650e−5	3.184e−5	2.781e−5	2.446e−5	2.157e−5
70000	4.970e−5	4.245e−5	3.640e−5	3.125e−5	2.697e−5	2.335e−5	2.031e−5	1.772e−5	1.553e−5	1.367e−5
80000	2.970e−6	2.375e−6	1.891e−6	1.509e−6	1.208e−6	9.647e−7	7.736e−7	6.219e−7	5.007e−7	4.044e−7
90000	3.872e−6	3.146e−6	2.564e−6	2.095e−6	1.720e−6	1.416e−6	1.171e−6	9.733e−7	8.124e−7	6.820e−7
100000	1.248e−5	1.051e−5	8.904e−6	7.568e−6	6.457e−6	5.534e−6	4.763e−6	4.117e−6	3.576e−6	3.125e−6

Table 3: Corresponding to Figure 3.

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